Supplements: Differentially Private Selection using Enhanced Sensitivity Concepts

This document provides the proofs and omitted definitions in the main paper. It also provides the detailed explanations of computation used in the experiments.

I. Proofs and Omitted Definitions

Theorem 1. *M_{ELP} satisfies ϵ-differential privacy.*

Proof. We let $\mathcal{R} = \{1, 2, \ldots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{ELP}(x) = 1] \leq e^{\epsilon} \cdot \Pr[M_{ELP}(y) = 1]$. We denote $Z_r = \mathrm{Lap}_r\left(\frac{EGS_{u,\mathcal{R}}}{\epsilon}\right)$.

$$\begin{aligned}
\Pr[M_{ELP}(x) = 1] &= \int_{v \in (-\infty,\infty)} \Pr[u(x,1) + Z_1 = v] \\
&\quad \cdot \Pr[u(x,2) + Z_2 \leq v] \\
&\quad \cdots \Pr[u(x,m) + Z_m \leq v] \\
&= \int_{v \in (-\infty,\infty)} \Pr[Z_1 = v - u(x,1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x,2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(x,m)].
\end{aligned}$$

$$\begin{aligned}
\Pr[M_{ELP}(y) = 1] &= \int_{v \in (-\infty,\infty)} \Pr[Z_1 = v - u(y,1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(y,2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(y,m)] \\
&\geq \int_{v \in (-\infty,\infty)} \Pr[Z_1 = v - u(y,1)] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x,2) - T] \\
&\quad \cdots \Pr[Z_m \leq v - u(x,m) - T] \\
&\quad \left(T = \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x,r') - u(y,r')|\right) \\
&= \int_{v \in (-\infty,\infty)} \Pr[Z_1 = v - u(y,1) + T] \\
&\quad \cdot \Pr[Z_2 \leq v - u(x,2)] \\
&\quad \cdots \Pr[Z_m \leq v - u(x,m)].
\end{aligned}$$

Here, $|u(x,1) - u(y,1) + T|$

$$\begin{aligned}
&= \left| u(x,1) - u(y,1) + \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x,r') - u(y,r')| \right| \\
&\leq |u(x,1) - u(y,1)| + \max_{r' \in \mathcal{R} \setminus \{1\}} |u(x,r') - u(y,r')| \\
&\leq \max_{r \in \mathcal{R}} \max_{x,y:d(x,y)=1} \Bigg(|u(x,r) - u(y,r)| \\
&\qquad + \max_{r' \in \mathcal{R} \setminus \{r\}} |u(x,r') - u(y,r')| \Bigg) = EGS_{u,\mathcal{R}}.
\end{aligned}$$

Therefore, $\forall v : \Pr[Z_1 = v - u(x,1)] \leq e^{\epsilon} \cdot \Pr[Z_1 = v - u(y,1) + T]$, and $\Pr[M_{ELP}(x) = 1] \leq e^{\epsilon} \cdot \Pr[M_{ELP}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{ELP}(x) = r] \leq e^{\epsilon} \cdot \Pr[M_{ELP}(y) = r]$ for all x and y such that $d(x, y) = 1$. □

Lemma 1. $EGS_{u,\mathcal{R}} \leq 2 \cdot GS_{u,\mathcal{R}}$.

Proof.

$$\begin{aligned}
EGS_{u,\mathcal{R}} &= \max_{r \in \mathcal{R}} \max_{x,y:d(x,y)=1} \Bigg(|u(x,r) - u(y,r)| \\
&\qquad + \max_{r' \in \mathcal{R} \setminus \{r\}} |u(x,r') - u(y,r')| \Bigg) \\
&\leq \max_{r \in \mathcal{R}} \max_{x,y:d(x,y)=1} \Bigg(\max_{r' \in \mathcal{R}} |u(x,r') - u(y,r')| \\
&\qquad + \max_{r' \in \mathcal{R}} |u(x,r') - u(y,r')| \Bigg) \\
&= 2 \cdot \max_{r \in \mathcal{R}} \max_{x,y:d(x,y)=1} |u(x,r) - u(y,r)| \\
&= 2 \cdot GS_{u,\mathcal{R}}.
\end{aligned}$$

□

Theorem 2. *The ESPS satisfies $\left(\left(k + \frac{|\mathcal{R}|}{2} \cdot l\right) \cdot \epsilon\right)$-differential privacy.*

Proof. We let $\mathcal{R} = \{1, 2, \ldots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{ESPS}(x) = 1] \leq \exp\left(\left(k + \frac{m}{2} \cdot l\right) \cdot \epsilon\right) \cdot \Pr[M_{ESPS}(y) = 1]$. We denote $S(x) = \max_{r \in \mathcal{R}} S(x, r)$.

$$\begin{aligned}
&\Pr[M_{ESPS}(x) = 1] \\
&= \int_{v \in (-\infty,\infty)} \Pr\left[u(x,1) + \frac{S(x)}{2\alpha'} \cdot Z_1 = v\right] \\
&\quad \cdot \Pr\left[u(x,2) + \frac{S(x)}{2\alpha'} \cdot Z_2 \leq v\right] \\
&\quad \cdots \Pr\left[u(x,m) + \frac{S(x)}{2\alpha'} \cdot Z_m \leq v\right] \\
&= \int_{v \in (-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(x,1))}{S(x)}\right] \\
&\quad \cdot \Pr\left[Z_2 \leq \frac{2\alpha'(v - u(x,2))}{S(x)}\right] \\
&\quad \cdots \Pr\left[Z_m \leq \frac{2\alpha'(v - u(x,m))}{S(x)}\right].
\end{aligned}$$

$$
\begin{aligned}
&\Pr[M_{ESPS}(y)=1]\\
&= \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v-u(y,1))}{S(y)}\right]\\
&\qquad \cdot \Pr\left[Z_2 \le \frac{2\alpha'(v-u(y,2))}{S(y)}\right]\\
&\qquad \cdots \Pr\left[Z_m \le \frac{2\alpha'(v-u(y,m))}{S(y)}\right]\\
&\ge \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v-u(y,1))}{S(y)}\right]\\
&\qquad \cdot \Pr\left[Z_2 \le \frac{2\alpha'(v-u(x,2)-T)}{S(y)}\right]\\
&\qquad \cdots \Pr\left[Z_m \le \frac{2\alpha'(v-u(x,m)-T)}{S(y)}\right]\\
&\left(T = \max_{r'\in\mathcal{R}\setminus\{1\}} |u(x,r')-u(y,r')|\right)\\
&= \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v-u(y,1)+T)}{S(y)}\right]\\
&\qquad \cdot \Pr\left[Z_2 \le \frac{2\alpha'(v-u(x,2))}{S(y)}\right]\\
&\qquad \cdots \Pr\left[Z_m \le \frac{2\alpha'(v-u(x,m))}{S(y)}\right].
\end{aligned}
$$

Here, for all $v \in (-\infty,\infty)$

$$
\begin{aligned}
\forall r' \in \mathcal{R}\setminus\{1\}:\quad & \Pr\left[Z_{r'} \le \frac{2\alpha'(v-u(x,r'))}{S(x)}\right]\\
&\le e^{\frac{l\epsilon}{2}} \cdot \Pr\left[Z_{r'} \le \frac{2\alpha'(v-u(x,r'))}{S(y)}\right]
\end{aligned}
$$

because

$$
\frac{S(x)}{S(y)} = \frac{\max_r S(x,r)}{\max_r S(y,r)} \le e^{\beta'} = e^{\beta(l\epsilon)}
$$

$$
\text{and}\quad \frac{S(x)}{S(y)} \ge e^{-\beta'} = e^{-\beta(l\epsilon)}.
$$

In addition,

$$
\begin{aligned}
&\Pr\left[Z_1 = \frac{2\alpha'(v-u(x,1))}{S(x)}\right]\\
&\le e^{k\epsilon} \cdot \Pr\left[Z_1 = \frac{2\alpha'(v-u(y,1)+T)}{S(x)}\right] \qquad (1)
\end{aligned}
$$

because

$$
\begin{aligned}
&\frac{|2\alpha'(u(x,1)-u(y,1)+T)|}{S(x)}\\
&= \frac{2\alpha'\left|u(x,1)-u(y,1)+\max_{r'\in\mathcal{R}\setminus\{1\}}|u(x,r')-u(y,r')|\right|}{S(x)}\\
&\le \frac{2\alpha'\left(|u(x,1)-u(y,1)|+\max_{r'\in\mathcal{R}\setminus\{1\}}|u(x,r')-u(y,r')|\right)}{S(x)}\\
&\le \frac{2\alpha'\cdot ELS_{u,\mathcal{R}}(x,1)}{S(x)} \le \frac{2\alpha'\cdot S(x,1)}{S(x)} \le 2\alpha' = \alpha(2k\epsilon),
\end{aligned}
$$

and

$$
(1) \le e^{k\epsilon}\cdot e^{\frac{l\epsilon}{2}} \cdot \Pr\left[Z_1 = \frac{2\alpha'(v-u(y,1)+T)}{S(y)}\right].
$$

Therefore, $\Pr[M_{ESPS}(x) = 1] \le \exp\left(\left(k+\frac{m}{2}\cdot l\right)\cdot\epsilon\right)\cdot \Pr[M_{ESPS}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R}$: $\Pr[M_{ESPS}(x) = r] \le \exp\left(\left(k+\frac{m}{2}\cdot l\right)\cdot\epsilon\right)\cdot\Pr[M_{ESPS}(y) = r]$ for all x and y such that $d(x,y) = 1$. □

Lemma 2. $\forall x \in D^n, \forall r \in \mathcal{R} : ELS_{u,\mathcal{R}}(x,r) \le 2\cdot \max_{r'\in\mathcal{R}} LS_{u,\mathcal{R}}(x,r')$.

Proof.

$$
\begin{aligned}
ELS_{u,\mathcal{R}}(x,r) &= \max_{y:d(x,y)=1}\Bigg(|u(x,r)-u(y,r)|\\
&\qquad + \max_{r'\in\mathcal{R}\setminus\{r\}}|u(x,r')-u(y,r')|\Bigg)\\
&\le 2\cdot\max_{r'\in\mathcal{R}}\max_{y:d(x,y)=1}|u(x,r')-u(y,r')|\\
&= 2\cdot\max_{r'\in\mathcal{R}} LS_{u,\mathcal{R}}(x,r').
\end{aligned}
$$

□

Theorem 3. *M_{EDLP} satisfies ϵ-differential privacy.*

Proof. We let $\mathcal{R} = \{1,2,\ldots,m\}$. We show that for all x and y such that $d(x,y) = 1$, $\Pr[M_{EDLP}(x) = 1] \le e^{\epsilon}\cdot\Pr[M_{EDLP}(y) = 1]$. We denote $Z_r = \mathrm{Lap}_r\left(\frac{EDGS_{u,\mathcal{R}}}{\epsilon}\right)$.

$$
\begin{aligned}
\Pr[M_{EDLP}(x)=1] &= \int_{v\in(-\infty,\infty)} \Pr[Z_1 = v-u(x,1)]\\
&\qquad \cdot\Pr[Z_2 \le v-u(x,2)]\\
&\qquad \cdots\Pr[Z_m \le v-u(x,m)].
\end{aligned}
$$

$$
\begin{aligned}
\Pr[M_{EDLP}(y)=1] &= \int_{v\in(-\infty,\infty)} \Pr[Z_1 = v-u(y,1)]\\
&\qquad \cdot\Pr[Z_2 \le v-u(y,2)]\\
&\qquad \cdots\Pr[Z_m \le v-u(y,m)]\\
&\ge \int_{v\in(-\infty,\infty)} \Pr[Z_1 = v-u(y,1)]\\
&\qquad \cdot\Pr[Z_2 \le v-u(x,2)-T']\\
&\qquad \cdots\Pr[Z_m \le v-u(x,m)-T']\\
&\left(T' = \max_{r'\in\mathcal{R}\setminus\{1\}}(u(y,r')-u(x,r'))\right)\\
&= \int_{v\in(-\infty,\infty)} \Pr[Z_1 = v-u(y,1)+T']\\
&\qquad \cdot\Pr[Z_2 \le v-u(x,2)]\\
&\qquad \cdots\Pr[Z_m \le v-u(x,m)].
\end{aligned}
$$

$$
\begin{aligned}
\text{Here, }\ &|u(x,1)-u(y,1)+T'|\\
&= \left|u(x,1)-u(y,1)+\max_{r'\in\mathcal{R}\setminus\{1\}}(u(y,r')-u(x,r'))\right|\\
&\le \max_{r\in\mathcal{R}}\max_{x,y:d(x,y)=1}\Bigg|(u(x,r)-u(y,r))\\
&\qquad + \max_{r'\in\mathcal{R}\setminus\{r\}}(u(y,r')-u(x,r'))\Bigg|\\
&= EDGS_{u,\mathcal{R}}.
\end{aligned}
$$

Therefore, $\forall v : \Pr[Z_1 = v-u(x,1)] \le e^{\epsilon}\cdot\Pr[Z_1 = v-u(y,1)+T']$, and $\Pr[M_{EDLP}(x) = 1] \le e^{\epsilon}\cdot\Pr[M_{EDLP}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{EDLP}(x) = r] \le e^{\epsilon}\cdot\Pr[M_{EDLP}(y) = r]$ for all x and y such that $d(x,y) = 1$. □

Definition. (β*-Smooth Upper Bound on EDLS for Private Selection*)
For $\beta > 0$*, a function* $S : D^n \times \mathcal{R} \to \mathbb{R}$ *is a* β*-smooth upper bound on the enhanced and direction-aware local sensitivity of a score function* $u : D^n \times \mathcal{R} \to \mathbb{R}$ *if it satisfies the following requirements for any* $r \in \mathcal{R}$*:*

$$\begin{aligned} \forall x \in D^n : \quad & S(x,r) \geq EDLS_{u,\mathcal{R}}(x,r); \\ \forall x,y \in D^n, d(x,y) = 1 : \quad & S(x,r) \leq e^{\beta} \cdot S(y,r). \end{aligned}$$

Definition. (*Enhanced and Direction-Aware* β*-Smooth Sensitivity for Private Selection*)
For $\beta > 0$*, the enhanced and direction-aware* β*-smooth sensitivity of a score function* $u : D^n \times \mathcal{R} \to \mathbb{R}$ *at* x *for* $r \in \mathcal{R}$ *is*

$$EDS^*_{u,\mathcal{R},\beta}(x,r) = \max_{y \in D^n} \left(EDLS_{u,\mathcal{R}}(y,r) \cdot e^{-\beta \cdot d(y,x)} \right).$$

Theorem 4. *The EDSPS satisfies* $\left(\left(k + \frac{|\mathcal{R}|}{2} \cdot l\right) \cdot \epsilon\right)$*-differential privacy.*

Proof. We let $\mathcal{R} = \{1, 2, \ldots, m\}$. We show that for all x and y such that $d(x,y) = 1$, $\Pr[M_{EDSPS}(x) = 1] \leq \exp\left(\left(k + \frac{m}{2} \cdot l\right) \cdot \epsilon\right) \cdot \Pr[M_{EDSPS}(y) = 1]$. We denote $S(x) = \max_{r \in \mathcal{R}} S(x,r)$.

$$\begin{aligned} & \Pr[M_{EDSPS}(x) = 1] \\ = & \int_{v \in (-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(x,1))}{S(x)}\right] \\ & \cdot \Pr\left[Z_2 \leq \frac{2\alpha'(v - u(x,2))}{S(x)}\right] \\ & \cdots \Pr\left[Z_m \leq \frac{2\alpha'(v - u(x,m))}{S(x)}\right]. \end{aligned}$$

$$\begin{aligned} & \Pr[M_{EDSPS}(y) = 1] \\ = & \int_{v \in (-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1))}{S(y)}\right] \\ & \cdot \Pr\left[Z_2 \leq \frac{2\alpha'(v - u(y,2))}{S(y)}\right] \\ & \cdots \Pr\left[Z_m \leq \frac{2\alpha'(v - u(y,m))}{S(y)}\right] \\ \geq & \int_{v \in (-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1))}{S(y)}\right] \\ & \cdot \Pr\left[Z_2 \leq \frac{2\alpha'(v - u(x,2) - T')}{S(y)}\right] \\ & \cdots \Pr\left[Z_m \leq \frac{2\alpha'(v - u(x,m) - T')}{S(y)}\right] \\ & \left(T' = \max_{r' \in \mathcal{R} \setminus \{1\}} (u(y,r') - u(x,r'))\right) \\ = & \int_{v \in (-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1) + T')}{S(y)}\right] \\ & \cdot \Pr\left[Z_2 \leq \frac{2\alpha'(v - u(x,2))}{S(y)}\right] \\ & \cdots \Pr\left[Z_m \leq \frac{2\alpha'(v - u(x,m))}{S(y)}\right]. \end{aligned}$$

Here, for all $v \in (-\infty, \infty)$

$$\begin{aligned} \forall r' \in \mathcal{R} \setminus \{1\} : \quad & \Pr\left[Z_{r'} \leq \frac{2\alpha'(v - u(x,r'))}{S(x)}\right] \\ & \leq e^{\frac{l\epsilon}{2}} \cdot \Pr\left[Z_{r'} \leq \frac{2\alpha'(v - u(x,r'))}{S(y)}\right]. \end{aligned}$$

In addition,

$$\begin{aligned} & \Pr\left[Z_1 = \frac{2\alpha'(v - u(x,1))}{S(x)}\right] \\ \leq & \; e^{k\epsilon} \cdot \Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1) + T')}{S(x)}\right] \end{aligned} \tag{2}$$

because

$$\begin{aligned} & \frac{|2\alpha'(u(x,1) - u(y,1) + T')|}{S(x)} \\ = & \frac{2\alpha' \left|u(x,1) - u(y,1) + \max_{r' \in \mathcal{R} \setminus \{1\}} (u(y,r') - u(x,r'))\right|}{S(x)} \\ \leq & \frac{2\alpha' \cdot EDLS_{u,\mathcal{R}}(x,1)}{S(x)} \leq \frac{2\alpha' \cdot S(x,1)}{S(x)} \leq 2\alpha' = \alpha(2k\epsilon), \end{aligned}$$

and

$$(2) \leq e^{k\epsilon} \cdot e^{\frac{l\epsilon}{2}} \cdot \Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1) + T')}{S(y)}\right].$$

Therefore, $\Pr[M_{EDSPS}(x) = 1] \leq \exp\left(\left(k + \frac{m}{2} \cdot l\right) \cdot \epsilon\right) \cdot \Pr[M_{EDSPS}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R} : \Pr[M_{EDSPS}(x) = r] \leq \exp\left(\left(k + \frac{m}{2} \cdot l\right) \cdot \epsilon\right) \cdot \Pr[M_{EDSPS}(y) = r]$ for all x and y such that $d(x,y) = 1$. $\square$

Theorem 5. *The SPS satisfies* $\left(k \cdot \epsilon + \max\left(\frac{l}{2} \cdot \epsilon, \, |\mathcal{R}| \cdot \beta'\right)\right)$*-differential privacy when* Z_r *are sampled from the one-sided distribution* g*.*

Proof. We let $\mathcal{R} = \{1, 2, \ldots, m\}$. We show that for all x and y such that $d(x,y) = 1$, $\Pr[M_{SPS}(x) = 1] \leq \exp\left(k \cdot \epsilon + \max\left(\frac{l}{2} \cdot \epsilon, \, m \cdot \beta'\right)\right) \cdot \Pr[M_{SPS}(y) = 1]$. We denote $S(x) = \max_{r \in \mathcal{R}} S(x,r)$.

$$\begin{aligned} & \Pr[M_{SPS}(x) = 1] \\ = & \int_{v \in (-\infty,\infty)} \Pr\left[Z_1 = \frac{\alpha'(v - u(x,1))}{S(x)}\right] \\ & \cdot \Pr\left[Z_2 \leq \frac{\alpha'(v - u(x,2))}{S(x)}\right] \\ & \cdots \Pr\left[Z_m \leq \frac{\alpha'(v - u(x,m))}{S(x)}\right]. \end{aligned}$$

$$\begin{aligned} & \Pr[M_{SPS}(y) = 1] \\ = & \int_{v \in (-\infty,\infty)} \Pr\left[Z_1 = \frac{\alpha'(v - u(y,1))}{S(y)}\right] \\ & \cdot \Pr\left[Z_2 \leq \frac{\alpha'(v - u(y,2))}{S(y)}\right] \\ & \cdots \Pr\left[Z_m \leq \frac{\alpha'(v - u(y,m))}{S(y)}\right] \end{aligned}$$

$$\geq \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{\alpha'(v-u(y,1))}{S(y)}\right] \cdot \Pr\left[Z_2 \leq \frac{\alpha'(v-u(x,2)-S(x))}{S(y)}\right] \cdots \Pr\left[Z_m \leq \frac{\alpha'(v-u(x,m)-S(x))}{S(y)}\right]$$
$$= \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{\alpha'(v-u(y,1)+S(x))}{S(y)}\right] \cdot \Pr\left[Z_2 \leq \frac{\alpha'(v-u(x,2))}{S(y)}\right] \cdots \Pr\left[Z_m \leq \frac{\alpha'(v-u(x,m))}{S(y)}\right].$$

(I) When $S(x) \geq S(y)$, for all $v \in (-\infty, \infty)$,

$$\forall r' \in \mathcal{R}\backslash\{1\}: \quad \Pr\left[Z_{r'} \leq \frac{\alpha'(v-u(x,r'))}{S(x)}\right] \leq \Pr\left[Z_{r'} \leq \frac{\alpha'(v-u(x,r'))}{S(y)}\right].$$

In addition,

$$\Pr\left[Z_1 = \frac{\alpha'(v-u(x,1))}{S(x)}\right] \leq e^{k\epsilon} \cdot \Pr\left[Z_1 = \frac{\alpha'(v-u(y,1)+S(x))}{S(x)}\right] \quad (3)$$

because

$$0 \leq \frac{\alpha'(u(x,1)-u(y,1)+S(x))}{S(x)} \leq \frac{\alpha' \cdot 2S(x)}{S(x)} = 2\alpha' = \alpha(2k\epsilon),$$

and

$$(3) \leq e^{k\epsilon} \cdot e^{\frac{l\epsilon}{2}} \cdot \Pr\left[Z_1 = \frac{\alpha'(v-u(y,1)+S(x))}{S(y)}\right].$$

Therefore, $\Pr[M_{SPS}(x)=1] \leq e^{k\epsilon+\frac{l\epsilon}{2}} \cdot \Pr[M_{SPS}(y)=1]$.
(II) When $S(y) > S(x)$, note that when $\lambda < 0$, $\ln\left(\frac{h(z)}{e^{\lambda}\cdot h(e^{\lambda}\cdot z)}\right) < |\lambda|$ for $z \geq 0$. Therefore, for all $v \in (-\infty, \infty)$,

$$\forall r' \in \mathcal{R}\backslash\{1\}: \quad \Pr\left[Z_{r'} \leq \frac{\alpha'(v-u(x,r'))}{S(x)}\right] \leq e^{\beta'} \cdot \Pr\left[Z_{r'} \leq \frac{\alpha'(v-u(x,r'))}{S(y)}\right].$$

In addition,

$$\Pr\left[Z_1 = \frac{\alpha'(v-u(x,1))}{S(x)}\right] \leq e^{k\epsilon} \cdot e^{\beta'} \cdot \Pr\left[Z_1 = \frac{\alpha'(v-u(y,1)+S(x))}{S(x)}\right].$$

Therefore, $\Pr[M_{SPS}(x)=1] \leq e^{k\epsilon+m\beta'} \cdot \Pr[M_{SPS}(y)=1]$. Consequently, $\Pr[M_{SPS}(x)=1] \leq \exp\big(k\cdot\epsilon + \max\big(\frac{l}{2}\cdot\epsilon,\, m\cdot\beta'\big)\cdot\epsilon\big) \cdot \Pr[M_{SPS}(y)=1]$. In a similar manner, $\forall r \in \mathcal{R}$: $\Pr[M_{SPS}(x)=r] \leq \exp\big(k\cdot\epsilon + \max\big(\frac{l}{2}\cdot\epsilon,\, m\cdot\beta'\big)\cdot\epsilon\big) \cdot \Pr[M_{SPS}(y)=r]$ for all x and y such that $d(x,y)=1$. □

Theorem 6. *M_{EDOEP} satisfies ϵ-differential privacy.*

Proof. We let $\mathcal{R} = \{1,2,\ldots,m\}$. We show that for all x and y such that $d(x,y)=1$, $\Pr[M_{EDOEP}(x)=1] \leq e^{\epsilon}\cdot\Pr[M_{EDOEP}(y)=1]$. We denote $Z_r = \mathrm{Expo}_r\left(\frac{\epsilon}{EDOGS_{u,\mathcal{R}}}\right)$.

$$\Pr[M_{EDOEP}(x)=1] = \int_{v\in(-\infty,\infty)} \Pr[Z_1 = v-u(x,1)] \cdot \Pr[Z_2 \leq v-u(x,2)] \cdots \Pr[Z_m \leq v-u(x,m)].$$

$$\begin{aligned}\Pr[M_{EDOEP}(y)=1] &= \int_{v\in(-\infty,\infty)} \Pr[Z_1 = v-u(y,1)] \cdot \Pr[Z_2 \leq v-u(y,2)] \cdots \Pr[Z_m \leq v-u(y,m)]\\ &\geq \int_{v\in(-\infty,\infty)} \Pr[Z_1 = v-u(y,1)] \cdot \Pr[Z_2 \leq v-u(x,2)-T''] \cdots \Pr[Z_m \leq v-u(x,m)-T'']\\ &\quad \left(T'' = \max_{r'\in\mathcal{R}}(u(y,r')-u(x,r'))\right)\\ &= \int_{v\in(-\infty,\infty)} \Pr[Z_1 = v-u(y,1)+T''] \cdot \Pr[Z_2 \leq v-u(x,2)] \cdots \Pr[Z_m \leq v-u(x,m)].\end{aligned}$$

$$\begin{aligned}\text{Here, } &EDOGS_{u,\mathcal{R}}\\ &\geq u(x,1)-u(y,1)+T''\\ &= -(u(y,1)-u(x,1)) + \max_{r'\in\mathcal{R}}(u(y,r')-u(x,r'))\\ &\geq -\max_{r\in\mathcal{R}}(u(y,r)-u(x,r)) + \max_{r'\in\mathcal{R}}(u(y,r')-u(x,r'))\\ &= 0.\end{aligned}$$

Therefore, $\forall v$: $\Pr[Z_1 = v-u(x,1)] \leq e^{\epsilon}\cdot\Pr[Z_1 = v-u(y,1)+T'']$, and $\Pr[M_{EDOEP}(x)=1] \leq e^{\epsilon}\cdot\Pr[M_{EDOEP}(y)=1]$. In a similar manner, $\forall r \in \mathcal{R}$: $\Pr[M_{EDOEP}(x)=r] \leq e^{\epsilon}\cdot\Pr[M_{EDOEP}(y)=r]$ for all x and y such that $d(x,y)=1$. □

Definition. (*β-Smooth Upper Bound on EDOLS for Private Selection*)
For $\beta > 0$, a function $S : D^n \times \mathcal{R} \to \mathbb{R}$ is a β-smooth upper bound on the enhanced and direction-aware local sensitivity for one-sided distribution cases of a score function $u : D^n \times \mathcal{R} \to \mathbb{R}$ if it satisfies the following requirements for any $r \in \mathcal{R}$:

$$\forall x \in D^n: \qquad S(x,r) \geq EDOLS_{u,\mathcal{R}}(x,r);$$
$$\forall x,y \in D^n, d(x,y)=1: \qquad S(x,r) \leq e^{\beta}\cdot S(y,r).$$

Definition. (*Enhanced and Direction-Aware β-Smooth Sensitivity for Private Selection with One-Sided Distribution*)
For $\beta > 0$, the enhanced and direction-aware β-smooth sensitivity for one-sided distribution cases of a score function $u : D^n \times \mathcal{R} \to \mathbb{R}$ at x for $r \in \mathcal{R}$ is

$$EDOS^*_{u,\mathcal{R},\beta}(x,r) = \max_{y\in D^n}\left(EDOLS_{u,\mathcal{R}}(y,r)\cdot e^{-\beta\cdot d(y,x)}\right).$$

Theorem 7. *The EDOSPS satisfies* $\left(k\cdot\epsilon+\max\left(\frac{l}{2}\cdot\epsilon, |\mathcal{R}|\cdot\beta'\right)\right)$*-differential privacy.*

Proof. We let $\mathcal{R} = \{1, 2, \ldots, m\}$. We show that for all x and y such that $d(x, y) = 1$, $\Pr[M_{EDOSPS}(x) = 1] \leq \exp\bigl(k\cdot\epsilon + \max\bigl(\frac{l}{2}\cdot\epsilon, m\cdot\beta'\bigr)\bigr)\cdot\Pr[M_{EDOSPS}(y) = 1]$. We denote $S(x) = \max_{r\in\mathcal{R}} S(x, r)$.

$$\begin{aligned}
&\Pr[M_{EDOSPS}(x) = 1]\\
&= \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(x,1))}{S(x)}\right]\\
&\qquad\cdot\Pr\left[Z_2 \leq \frac{2\alpha'(v - u(x,2))}{S(x)}\right]\\
&\qquad\cdots\Pr\left[Z_m \leq \frac{2\alpha'(v - u(x,m))}{S(x)}\right].
\end{aligned}$$

$$\begin{aligned}
&\Pr[M_{EDOSPS}(y) = 1]\\
&= \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1))}{S(y)}\right]\\
&\qquad\cdot\Pr\left[Z_2 \leq \frac{2\alpha'(v - u(y,2))}{S(y)}\right]\\
&\qquad\cdots\Pr\left[Z_m \leq \frac{2\alpha'(v - u(y,m))}{S(y)}\right]\\
&\geq \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1))}{S(y)}\right]\\
&\qquad\cdot\Pr\left[Z_2 \leq \frac{2\alpha'(v - u(x,2) - T'')}{S(y)}\right]\\
&\qquad\cdots\Pr\left[Z_m \leq \frac{2\alpha'(v - u(x,m) - T'')}{S(y)}\right]\\
&\left(T'' = \max_{r'\in\mathcal{R}}(u(y,r') - u(x,r'))\right)\\
&= \int_{v\in(-\infty,\infty)} \Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1) + T'')}{S(y)}\right]\\
&\qquad\cdot\Pr\left[Z_2 \leq \frac{2\alpha'(v - u(x,2))}{S(y)}\right]\\
&\qquad\cdots\Pr\left[Z_m \leq \frac{2\alpha'(v - u(x,m))}{S(y)}\right].
\end{aligned}$$

(I) When $S(x) \geq S(y)$, for all $v \in (-\infty, \infty)$,

$$\begin{aligned}
\forall r' \in \mathcal{R}\backslash\{1\}: \quad &\Pr\left[Z_{r'} \leq \frac{2\alpha'(v - u(x,r'))}{S(x)}\right]\\
&\leq \Pr\left[Z_{r'} \leq \frac{2\alpha'(v - u(x,r'))}{S(y)}\right].
\end{aligned}$$

In addition,

$$\begin{aligned}
&\Pr\left[Z_1 = \frac{2\alpha'(v - u(x,1))}{S(x)}\right]\\
&\leq e^{k\epsilon}\cdot\Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1) + T'')}{S(x)}\right]
\end{aligned}\tag{4}$$

because

$$\begin{aligned}
0 \leq \frac{2\alpha'(u(x,1) - u(y,1) + T'')}{S(x)} &\leq \frac{2\alpha'\cdot EDOLS_{u,\mathcal{R}}}{S(x)}\\
&\leq 2\alpha' = \alpha(2k\epsilon),
\end{aligned}$$

and (4) $\leq e^{k\epsilon}\cdot e^{\frac{l\epsilon}{2}}\cdot\Pr\left[Z_1 = \dfrac{2\alpha'(v - u(y,1) + T'')}{S(y)}\right]$.

Therefore, $\Pr[M_{EDOSPS}(x) = 1] \leq e^{k\epsilon+\frac{l\epsilon}{2}}\cdot\Pr[M_{EDOSPS}(y) = 1]$.

(II) When $S(y) > S(x)$, note that when $\lambda < 0$, $\ln\left(\frac{h(z)}{e^{\lambda}\cdot h(e^{\lambda}\cdot z)}\right) < |\lambda|$ for $z \geq 0$. Therefore, for all $v \in (-\infty, \infty)$,

$$\begin{aligned}
\forall r' \in \mathcal{R}\backslash\{1\}: \quad &\Pr\left[Z_{r'} \leq \frac{2\alpha'(v - u(x,r'))}{S(x)}\right]\\
&\leq e^{\beta'}\cdot\Pr\left[Z_{r'} \leq \frac{2\alpha'(v - u(x,r'))}{S(y)}\right].
\end{aligned}$$

In addition,

$$\begin{aligned}
&\Pr\left[Z_1 = \frac{2\alpha'(v - u(x,1))}{S(x)}\right]\\
&\leq e^{k\epsilon}\cdot e^{\beta'}\cdot\Pr\left[Z_1 = \frac{2\alpha'(v - u(y,1) + T'')}{S(x)}\right].
\end{aligned}$$

Therefore, $\Pr[M_{EDOSPS}(x) = 1] \leq e^{k\epsilon+m\beta'}\cdot\Pr[M_{EDOSPS}(y) = 1]$.

Consequently, $\Pr[M_{EDOSPS}(x) = 1] \leq \exp\bigl(k\cdot\epsilon+\max\bigl(\frac{l}{2}\cdot\epsilon, m\cdot\beta'\bigr)\cdot\epsilon\bigr)\cdot\Pr[M_{EDOSPS}(y) = 1]$. In a similar manner, $\forall r \in \mathcal{R}: \Pr[M_{EDOSPS}(x) = r] \leq \exp\bigl(k\cdot\epsilon + \max\bigl(\frac{l}{2}\cdot\epsilon, m\cdot\beta'\bigr)\cdot\epsilon\bigr)\cdot\Pr[M_{EDOSPS}(y) = r]$ for all x and y such that $d(x, y) = 1$. ☐

Theorem 8. *Given a set of datasets $U \subseteq D^n$, for any $\beta\,(> 0)$ satisfying*

$$\beta \leq \min_{x\notin U} \frac{1}{ud(x)}\cdot\ln\left(\frac{GS_f}{LS_f(x)}\right), \tag{5}$$

where $ud(x) := \min_{y\in U} d(y, x)$ represents the shortest distance between x and U, the following function S is a β-smooth upper bound:

$$S(x) = GS_f\cdot e^{-\beta\cdot ud(x)}.$$

Proof. From (5), for all $x \notin U$,

$$\beta \leq \frac{1}{ud(x)}\cdot\ln\left(\frac{GS_f}{LS_f(x)}\right) \iff LS_f(x) \leq GS_f\cdot e^{-\beta\cdot ud(x)}.$$

Therefore, it is sufficient to show that $\forall x, y, d(x, y) = 1: S(x) \leq e^{\beta}\cdot S(y)$.

(I) When $x, y \in U$, $S(x) = S(y) = GS_f$.

(II) When $x \in U$ and $y \notin U$, $S(x) = GS_f$ and $S(y) = GS_f\cdot e^{-\beta}$ because $ud(y) = d(x, y) = 1$.

(III) When $x \notin U$ and $y \in U$, $S(x) = GS_f\cdot e^{-\beta}$ and $S(y) = GS_f$.

(IV) When $x \notin U$ and $y \notin U$, where X satisfies $X \in U$ and $d(X, x) = ud(x)$, and Y satisfies $Y \in U$ and $d(Y, y) = ud(y)$,

$$\begin{aligned}
e^{\beta}\cdot S(y) &= e^{\beta}\cdot GS_f\cdot e^{-\beta\cdot d(Y,y)}\\
&\geq e^{\beta}\cdot GS_f\cdot e^{-\beta\cdot d(X,y)}\ [\because d(Y,y) \leq d(X,y)]\\
&\geq e^{\beta}\cdot GS_f\cdot e^{-\beta\cdot(d(X,x)+d(x,y))}\\
&\qquad[\because d(X,y) \leq d(X,x) + d(x,y)]\\
&= GS_f\cdot e^{-\beta\cdot d(X,x)} = S(x).
\end{aligned}$$

☐

II. Computation in Experiments

A. Computation of $d(x, y)$

The Hamming distance between two datasets was approximated based on the values of b and c as follows:

$$
\begin{aligned}
& d(T(b,c),\, T(b',c')) \\
= \quad & \begin{cases} \left\lceil \frac{|(b+c)-(b'+c')|}{2} \right\rceil & ((b-b')\cdot(c-c') \geq 0) \\ \left\lceil \frac{\max\{|b-b'|,\, |c-c'|\}}{2} \right\rceil & ((b-b')\cdot(c-c') < 0) \end{cases},
\end{aligned}
$$

where $T(b, c)$ represents a table that can be formed as the following table:

		Non-Transmitted Allele A_1	Non-Transmitted Allele A_2	Total
Transmitted Allele	A_1	a	b	$a+b$
Transmitted Allele	A_2	c	d	$c+d$
Total		$a+c$	$b+d$	$2n$

B. Computation of $ud(x)$

Theorem 8 was used to compute $S(x, r)$ for the proposed mechanisms using smooth sensitivity, M_{ESPS}, M_{EDSPS}, and M_{EDOSPS}. For these cases, U was set as $\{y \,|\, \exists r' \in \mathcal{R}\backslash\{r\}.\, \max_{z:d(y,z)=1}(|u(y,r) - u(z,r)| + |u(y,r') - u(z,r')|) > 12\}$, $\{y \,|\, \exists r' \in \mathcal{R}\backslash\{r\}.\, \max_{z:d(y,z)=1} |(u(y,r) - u(z,r)) + (u(z,r') - u(y,r'))| > 12\}$, and $\{y \,|\, \exists r' \in \mathcal{R}.\, \max_{z:d(y,z)=1}((u(y,r) - u(z,r)) + (u(z,r') - u(y,r'))) > 12\}$, respectively. Here, it was assumed that the maximum changes in the values of $\max_{y:d(x,y)=1} |u(y,r) - u(x,r)|$, $\max_{y:d(x,y)=1}(u(y,r) - u(x,r))$, and $\max_{y:d(x,y)=1}(u(x,r) - u(y,r))$ when x was changed to z s.t. $d(z,x) = 1$ were almost $\frac{32}{b_r+c_r}$. Then, the value of $ud(x)$ for $x \notin U$ was approximated to

$$
\begin{aligned}
& \min_{r'} \left\lceil \frac{(12 - (ls_r + ls_{r'}))\,(b_r + c_r)(b_{r'} + c_{r'})}{32 \cdot (b_r + c_r + b_{r'} + c_{r'})} + 1 \right\rceil, \\
& \min_{r'} \left\lceil \frac{(12 - (ls1_r + ls2_{r'}))\,(b_r + c_r)(b_{r'} + c_{r'})}{32 \cdot (b_r + c_r + b_{r'} + c_{r'})} + 1 \right\rceil, \\
\text{and } & \min_{r'} \left\lceil \frac{(12 - (ls1_r + ls2_{r'}))\,(b_r + c_r)(b_{r'} + c_{r'})}{32 \cdot (b_r + c_r + b_{r'} + c_{r'})} + 1 \right\rceil,
\end{aligned}
$$

respectively, where $ls_r = \max_{y:d(x,y)=1} |u(x,r) - u(y,r)|$, $ls1_r = \max_{y:d(x,y)=1}(u(x,r) - u(y,r))$, and $ls2_r = \max_{y:d(x,y)=1}(u(y,r) - u(x,r))$.